Michael **Schlichtkrull**

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About Me

I am a lecturer (≈ assistant professor) at Queen Mary University of London, working on natural language processing. I am especially interested in evidential reasoning and factuality using LLMs and other NLP approaches. My research interests and publication history span automated reasoning about truth and trustworthiness with large language models, knowledge-intensive NLP, graph neural networks, explainable AI, and evaluation of AI systems, on all of which I have published in top venues for machine learning or NLP.

I focus on machine learning and natural language processing as applied to large textual and structured data sources, such as Wikipedia, knowledge bases, or the open web. I study the foundational challenge of extracting insights from knowledge bases that include *untrustworthy* sources, allowing agents to extract reliable knowledge even in the presence of mis- and disinformation. I develop explainable, feasible, and epistemically responsible processes for such data, learning from human experts, such as journalists and historians.

Work Experience

Lecturer in Computer Science (Teaching & Research)
London, United Kingdom

Queen Mary University of London *January 2024 - Present*

- Full-time academic position at Queen Mary, University of London (≈ Assistant Professor).
- Academic research on automated media literacy, automated fact-checking, interpretability for LLMs, evaluating reasoning behaviour, and applications of LLMs to medical and legal texts.
- Work involving multiple modalities, including text, images, sign language clips, tables, and graphs.
- Module organiser for the courses EECS700P & IOT7005P on Risk and Decision Making with Artificial Intelligence.
- Supervising students at PhD, MSc, and undergraduate level.
- Senior area chair at *ACL-venues, coorganiser of the FEVER workshop and shared task.

Postdoctoral Research Associate & Affiliated Lecturer
Cambridge, United Kingdom

University of Cambridge *March 2021 - January 2024*

- Research position at the University of Cambridge, working with Andreas Vlachos on automated fact-checking.
- Worked on a breadth of problems including dataset collection, evaluation, modelling, and epistemology.
- Multiple publications at top venues, including NeurIPS, ACL, TACL, and EMNLP.
- Created data for and organised the AVeriTeC shared task at FEVER 2024, evaluating LLMs and other NLP systems on fact-checking real-world misinformation. With 21 submissions, the shared task exceeded expectations.
- Cosupervised students at PhD, MPhil, and undergraduate level.
- Worked as a co-organizer for the FEVER 2021, 2022, and 2023 workshops at EMNLP 2021, ACL 2022, and EACL 2023 respectively.

Research Intern
London, United Kingdom

Facebook AI Research *June 2020 - September 2020*

- Research internship in NLP at FAIR London, working with Sebastian Riedel, Mike Lewis, and Scott Yih.
- Development of the first system for fact-checking capable of both retrieving and reasoning with tabular data, published at ACL 2021.

Applied Science Intern
Berlin, Germany

Amazon *July 2018 - September 2018*

- Research internship in NLP at Amazon Berlin, working mainly with Weiwei Cheng and Alex Klementiev.
- Development of systems and evaluation practices for diverse question generation over natural text.

Student Assistant in IT Development
Copenhagen, Denmark

Saxo.com ApS *June 2014 - June 2016*

- Development of server-side software for online booksales, working mainly in C# and SQL.
- Implementation and exposure of functionalities through a public API, as well as occasional front-end work.

Teaching Assistant for Algorithms and Data Structures I & II
Copenhagen, Denmark

Technical University of Denmark *February 2012 - December 2012*

- Exercise sessions and homework grading for Algorithms and Data Structures I, taught by Inge Li Gøtz. Introductory material including sorting, searching, heaps, trees, and elementary graph theory.
- Exercise sessions for Algorithms and Data Structures II, taught by Paul Fischer. Higher-level material including flow algorithms, balanced data structures, and graph partitioning.

Education

PhD in Computer Science

Amsterdam, Netherlands & Edinburgh, United Kingdom

University of Amsterdam — *Sep 2016 - June 2021*

- PhD topic was the use of graph-neural networks to model structured data for natural language processing. Advised by Ivan Titov.
- Introduced the first graph neural network models for relational data, resulting in a highly cited publication (6000+) on link prediction.
- Publications to top conferences in AI and NLP, including ACL, ICLR, EACL, and ESWC.
- Developed interpretability techniques for GNNs, now included as standard in libraries such as PyTorch Geometric and DIG.

Master of Science in IT and Cognition

Copenhagen, Denmark

University of Copenhagen — *Sep 2014 - Aug 2016*

- Thesis subject was cross-lingual dependency parsing for truly low-resource languages, supervised by Anders Søgaard. My work resulted in state-of-the-art performance as well as a publication at EACL.
- Courses in natural language processing, statistical machine learning, probabilistic modelling, linguistics, and image processing.

Bachelor of Science in Software Engineering

Copenhagen, Denmark

Technical University of Denmark — *Sep 2010 - Jul 2013*

- Thesis subject was coreference resolution in semantic parsing, using formal logic. My supervisor was Jørgen Villadsen.
- Courses in mathematics, probability theory, graph theory, algorithms, formal logic, and computer science modelling.

Selected Publications

Attacks by Content: Automated Fact-checking is an AI Security Issue in which I demonstrate that LLM-based agents are vulnerable to indirect persuasion attacks when given access to retrieval tools, and show that automated fact-checking and source critical reasoning are effective countermeasures. Schlichtkrull, M. In *EMNLP, 2025*.

PledgeTracker: A System for Monitoring the Fulfilment of Pledges in which we deploy an automated fact-checking system to the real-world use case of tracking political pledges, powering `https://fullfact.org/government-tracker/`. Chen, Y., Schlichtkrull, M., Deng, Z., Corney, D., Asl, N., Salisbury, J., Dudfield, A., Vlachos, A. In *EMNLP System Demonstrations, 2025*.

Generating Media Background Checks for Automated Source Critical Reasoning in which I identify source trustworthiness as a key vulnerability of retrieval-augmented models and LLM-based agents, and propose a new dataset and baseline for automatically finding indicators of reliability. Schlichtkrull, M. In *Findings of EMNLP, 2024*.

AVeriTeC: A Dataset for Real-world Claim Verification with Evidence from the Web in which we introduce a new high-quality dataset for the development and evaluation of automated fact-checking systems, reverse-engineering the processes used by journalistic fact-checkers to gather evidence and judge veracity. Schlichtkrull, M., Guo, Z., and Vlachos, A. In *NeurIPS, 2023*.

The Intended Uses of Automated Fact-Checking Artefacts: Why, How and Who in which we study the intentions of researchers and designers of automated fact-checking systems, arguing that current practise is dominated by vague narratives that hold back the technology. Schlichtkrull, M, Ousidhoum, N., and Vlachos, A. In *Findings of EMNLP, 2023*.

A Survey on Automated Fact-checking in which we survey research directions and recent developments in automated fact-checking, highlight the connections between various subtasks, and discuss future challenges and research directions. The curated GitHub page is a highly used community resource. Guo, Z., Schlichtkrull, M., and Vlachos, A. In *TACL 2022*, presented at *ACL 2022*.

FEVEROUS: Fact Extraction and VERification Over Unstructured and Structured information in which we introduce the FEVEROUS dataset for fact-checking over structured and unstructured evidence, a new benchmark requiring modelling of both text and tables. Aly, R., Guo, Z., Schlichtkrull, M., Thorne, J., Vlachos, A., Christodoulopoulos, C., Cocarascu, O., and Mittal, A. In *NeurIPS 2021*.

Joint Verification and Reranking for Open Fact Checking Over Tables in which we investigate fact-checking over large collections of tables, producing the first results in the open domain along with a state-of-the-art attention-based model for both the open and the closed domain. Schlichtkrull, M., Karpukhin, V., Oğuz, B., Lewis, M., Yih, W., and Riedel, S. In *ACL 2021*.

Interpreting Graph Neural Networks for NLP With Differentiable Edge Masking in which we propose a technique for interpreting graph neural networks by identifying relevant and superfluous edges through a learned erasure function, and apply said technique to study models NLP tasks. Schlichtkrull, M., De Cao, N., and Titov, I. Spotlight paper at *ICLR 2021*.

Modeling Relational Data with Graph Convolutional Networks in which we obtain state-of-the-art results with for relational link prediction by developing the first graph neural network for directed, labeled graphs. Schlichtkrull, M., Kipf, T. N., Bloem, P., Berg, R.

V. D., Titov, I., & Welling, M. In *ESWC, 2018*. Won best student research paper award.

Cross-Lingual Dependency Parsing with Late Decoding for Truly Low-Resource Languages in which we set a new state of the art in dependency parsing for low-resource languages using annotation project and LSTMs. Schlichtkrull, M., Søgaard, A. In *EACL, 2017*.

Grants

AdSolve Co-investigator with collaborators from Queen Mary University of London, University of Sheffield, and University of Warwick on the Responsible AI UK Keystone grant AdSolve, where we have been awarded £4.38 million for addressing the sociotechnical limitations of large language models.

FEVER-IT Co-investigator with collaborators from Cambridge on the Alan Turing Institute / DSO National Laboratories Partnership grant FEVER-IT, where we have been awarded £350,000 for the modelling of multimodal misinformation.

Supervision & Teaching

EECS7005P & IOT7005P Risk and Decision Making with Artificial Intelligence I am currently module organizer for the undergraduate and apprenticeship versions of this course at Queen Mary, University of London. The course covers an introduction to probability and Bayes' theorem, reasoning with Bayesian networks and influence diagram, and a part on causal reasoning.

L90 Natural Language Processing Co-taught with Weiwei Sun at the University of Cambridge. The course covers a general introduction to NLP, intended for third-year undergraduates as well as MPhil students. My part was on generative models, focusing on summarization.

Guneet Singh Kohli I am currently co-supervising Guneet Singh Kohli's PhD research at QMUL together with Maria Liakata. His topic is evaluation of reasoning processes in LLMs, focusing on reference-free evaluation.

Eric Chamoun I am currently co-supervising Eric Chamoun's PhD research at Cambridge together with Andreas Vlachos. His topic is automated fact-checking for scientific claims, focusing on the use of peer reviews as data for fact-checking models. Our work has resulted in publications at ACL 2024, and EMNLP 2025.

Mubashara Akhtar I co-supervised Mubashara Akhtar's project as a visiting PhD researcher at Cambridge, together with Andreas Vlachos. Her topic was evaluation of evidence retrieval for automated fact-checking. Our work has resulted in a publication accepted to TACL.

Postgraduate students @ QMUL I am currently supervising six postgraduate dissertations at Queen Mary. Topics include automated fact-checking for Arabic data, mitigating gullibility in AI agents, long-context language modeling, evidence search in automated fact-checking, LLMs for heart disease prediction, and automated analysis of sportswashing discussions with NLP.

Undergraduate students @ QMUL I am currently supervising four undergraduate dissertations at Queen Mary. Topics include mitigating hallucinations in LLMs, tool-use in LLMs, finance-domain fact-checking, and machine translation for sign language.

Postgraduate students @ Cambridge I cosupervised two postgraduate dissertations in Cambridge. Topics included modeling lexical semantics with graph neural networks, and editing knowledge in LLMs to improve factuality. The former resulted in a publication at EMNLP 2023.

Undergraduate students @ Cambridge In 2022 and 2023 I co-supervised two undergraduate dissertations at Cambridge. Topics included automatically generated adversarial attacks against fact-checking models, and fact-checking over tables.

Pembroke Summer School I participated as a superviser in the summer school at Pembroke College, Cambridge, supervising undergraduate students for a six week period. Projects I supervised included graph neural networks for autonomous public transport, and large language models for education.

Skills

Natural languages	English (fluent), Danish (native), German (Proficient), Spanish (Intermediate), Mandarin (Beginner)
Programming languages	Python, Java, C#, C, C++, ML, Prolog, F#, Matlab, Javascript, Visual BASIC
Frameworks	Pytorch, Tensorflow, Theano, Huggingface Transformers, Scikit-learn, NumPy, Pandas
Other technologies	React, Express, Node.js, SQL, MongoDB, SPARQL, CSS, ASP.NET